Practical 1

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import pandas as pd
import numpy as np

# --- 3. Load the Dataset ---
df = pd.read_csv('Iris Dataset Prac (1,3.2,6,10).csv')

# --- 4. Data Preprocessing ---
print("1. Dimensions of Data:", df.shape)
print("\n2. Missing Values Check:\n", df.isnull().sum())
print("\n3. Initial Statistics:\n", df.describe())

# --- 5. Data Formatting & Normalization ---
print("\n4. Variable Types Before Conversion:\n", df.dtypes)

# Example Type Conversion (Ensuring Id is string if needed, or numeric)
df['Id'] = df['Id'].astype(int)

# Data Normalization (Min-Max Scaling for numeric columns)
# Formula: (val - min) / (max - min)
numeric_cols = ['SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm']
for col in numeric_cols:
    df[col] = (df[col] - df[col].min()) / (df[col].max() - df[col].min())

print("\n5. Data after Normalization (First 5 rows):\n", df.head())

# --- 6. Turn Categorical into Quantitative ---
# Converting 'Species' into numeric dummy variables
df_encoded = pd.get_dummies(df, columns=['Species'])
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print("\n6. Final Quantitative Data (First 5 rows):\n", df_encoded.head())
```